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ORIGINAL RESEARCH ARTICLE

Re-investigation of the Phase Equilibria and Thermodynamic Assessment of the Mg-Sm Binary System

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Abstract The Mg-Sm phase diagram has been experimentally studied by x-ray diffraction, scanning electron microscope equipped with energy dispersive spectrometer, and differential scanning calorimetry. Five binary compounds, $Mg_{41}Sm_5$, Mg_5Sm, Mg_3Sm, Mg_2Sm and MgSm were confirmed to exist in the Mg-Sm system. In addition to Mg_2Sm and MgSm, Mg_3Sm was found to be a congruently melting phase. The Mg_5Sm and Mg_2Sm were found to be only stable at high temperatures and decompose above 400 °C in this work. A special effort was made to determine the extent of the solid solubility ranges of Mg_3Sm, Mg_2Sm, MgSm and, γ-Sm. The Mg-Sm binary system was modeled using the Calphad approach based on new experimental data of this work and all reliable experimental information from literature. A complete thermodynamic description of the Mg-Sm system is obtained and extensive comparisons between calculated and experimental data are presented, indicating that almost all available experimental and theoretical data are fitted satisfactorily.

Keywords experimental phase diagram · Mg-Sm binary system · thermodynamic assessment

1 Introduction

Magnesium alloys exhibit exceptional physical and chemical properties, including a low modulus of elasticity, low density, high specific strength, strong damping characteristics, good biocompatibility, and a substantial hydrogen storage capacity. These attributes render magnesium alloys highly promising for applications in aerospace, 3C electronics products (Computers, Communications and Consumer electronics), new energy technologies, biomedicine, and more.[1–4]

Since the 21st century, Mg-RE (Rare Earth) binary alloys have been widely studied, revealing that rare earth elements can enhance the properties of magnesium alloys.[5] Samarium (Sm), being one of the light rare earth elements, exhibits significant potential owing to its abundant reserves. Research has suggested that the addition of Sm to magnesium alloys results in grain refinement,[6,7] enhanced corrosion resistance,[8,9] solution strengthening, aging strengthening,[10,11] and improved hydrogen storage performance.[12,13] Furthermore, Mg-1Sm alloy, characterized by good biocompatibility and 40.9% ductility, can served as a medical material, making it an optimal choice for cardiovascular stents.[14]

Accurate phase diagrams are paramount in alloy design. The motivation for our experimental reinvestigation on the Mg-Sm binary system phase diagram arises from our

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ongoing experimental research on the isothermal section of the Mg-Sn-Sm ternary system at 400 °C. In this study of the ternary system, we observed the apparent decomposition of Mg_5Sm and Mg_2Sm at temperatures exceeding 400 °C, which contradicts the widely accepted experimental phase diagram of the Mg-Sm system (Fig. 1) as reported by Saccone et al.[15]

Therefore, the objectives of this study are: (i) to experimentally reinvestigate the Mg-Sm phase diagram, specifically focusing on elucidating the formation modes and stability temperature ranges of Mg_5Sm and Mg_2Sm, and (ii) to conduct a re-assessment of the thermodynamic description of the Mg-Sm binary system based on the updated experimental phase diagram.

2 Literature Information

2.1 Phase Diagram

Iandelli and Palenzona[16] found that the crystal structure of MgSm is of the CsCl type. Kinzhibalo et al.,[17] through x-ray diffraction investigation, determined that the $Mg_{41}Sm_5$ phase has a tetragonal lattice. Rokhlin et al.,[18] utilizing electric resistance measurement and microscopic analysis, identified a eutectic reaction occurring at 542 °C on the Mg side. They determined that the maximum solid solubility of Sm in Mg is 0.99 at.% Sm at the eutectic transition point, while at 200 °C, the solid solubility of Sm in Mg is 0.063 at.% Sm. Rokhlin[19] established the Mg-Sm phase diagram

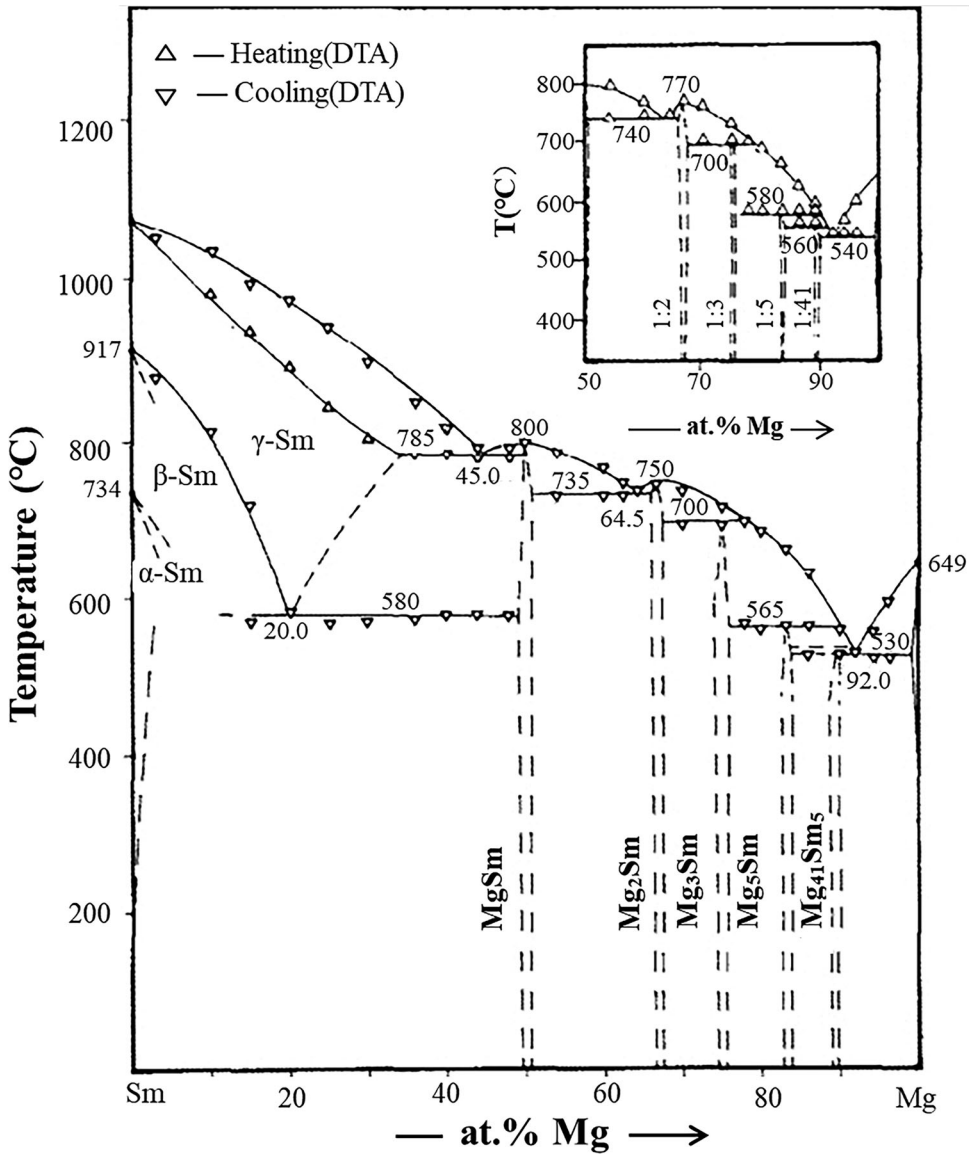


Fig. 1 The Mg-Sm binary phase diagram reported by Saccone et al.[[15]]

in the range of 0 ~ 11.68 at.% Sm through thermal analysis and microanalysis. Zheng et al.,[20] using X-ray diffraction, differential thermal analysis, and electron probe, determined the Mg-Sm binary phase diagram, comprising four binary phases: MgSm, Mg_2Sm, Mg_3Sm and $Mg_{13}Sm_2$. The latest experimental phase diagram of Mg-Sm is derived from the study of Saccone et al.,[15] who applied differential thermal analysis (DTA), x-ray diffraction (XRD), metallographic and microprobe analysis. In contrast to the phase diagram proposed by Zheng et al.,[20] Saccone et al.[15] suggested that the binary phase $Mg_{13}Sm_2$ at the Mg-rich side does not exist and should be replaced by two binary phases, Mg_5Sm and $Mg_{41}Sm_5$. Figure 1 illustrates the phase diagram proposed by Saccone et al.,[15] with the liquid phase and invariant equilibrium temperatures primarily based on the cooling effects of DTA. They also provided a partial phase diagram of Mg-Sm on the Mg-rich side based on DTA data from the heating curves. Typically, the thermal effect temperature of the cooling curve will be lower than the corresponding thermal effect temperature of the heating curve due to a certain degree of undercooling required for solidification, and the data of Saccone et al.[15] are consistent in this regard. Five intermetallic compounds, namely $Mg_{41}Sm_5$, Mg_5Sm, Mg_3Sm, Mg_2Sm and MgSm, are reported to exist in the binary system, with Mg_2Sm and MgSm being congruently melting phases. The solid solubility of each phase, as indicated by the dashed lines, is based on earlier literature data.[18,21] The solid solubility of Sm allotropes was not determined and the phase relationships among of Sm allotropes remain uncertain.

The crystallographic data for the reported solid phases in Mg-Sm binary systems is presented in Table 1.

2.2 Thermodynamic Data

Thermodynamic data regarding the Mg-Sm binary system are rather limited. Cacciamani et al.[28] reported the formation enthalpy of Mg-Sm alloys using calorimetric measurement. Delsante and Borzone[29] determined the formation enthalpy of $Mg_{41}Sm_5$ (− 5.5 ± 2 kJ/mol), Mg_5Sm (− 8.5 ± 2 kJ/mol), Mg_3Sm (− 12.0 ± 2 kJ/mol), Mg_2Sm (− 16.0 ± 2 kJ/mol) and MgSm (− 18.5 ± 2 kJ/mol) at 300 K through high-temperature direct synthesis calorimetry.

Tao et al.[30] calculated the formation enthalpy of $Mg_{41}Sm_5$, Mg_3Sm, Mg_2Sm, and MgSm using first principles, while Kang et al.[31] employed both first principles and Miedema's model to calculated the formation enthalpy of $Mg_{41}Sm_5$, Mg_3Sm, Mg_2Sm, and MgSm. However, there is a considerable difference between the calculated values of Kang et al.[31] and Tao et al.[30] and the experimental results of Cacciamani et al.[[28]] and Delsante and Borzone.[29]

Jia et al.[32] and Kang et al.[31] performed thermodynamic assessments of the Mg-Sm systems. The thermodynamic optimization by Jia et al.[32] was performed by Thermo-Calc software and the liquid phase was simulated using Bragg-Williams random mixing model. Kang et al.,[31] on the other hand, adopted Modified Quasi-chemical Model for liquid phase with FactSage software.

3 Experimental Work

3.1 Experimental Procedure

The initial materials were ingots of the pure elements (Mg, purity: 99.99%; Sm purity: 99.5%). The raw material was placed in a corundum crucible and melted with a high-frequency induction furnace. To prevent oxidation during the sample synthesis, the melting was conducted in a sealed quartz tube that was first evacuated and then filled with high-purity argon gas. The samples were held in the liquid state for about two minutes to promote homogenization of the sample composition. Each sample weighs 3 ~ 5

Table 1 Crystal structure data for the solid phases in the Mg-Sm system and the thermodynamic model used in present description.

Phase	Pearson symbol	Prototype	Space group	Thermodynamic model used in present description	References
Liquid	…	…	…	(Mg,Sm)	…
Mg	*hP*2	Mg	*P6₃/mmc*	HCP_A3: $(Mg,Sm)_1VA_{0.5}$	[15,22,23]
$Mg_{41}Sm_5$	*tI*92	Ce_5Mg_{41}	*I4/m*	$Mg_{0.8913}Sm_{0.1087}$	[15,17]
Mg_5Sm	*cF*440	$GdMg_5$	*F-4₃ m*	$Mg_{0.8333}Sm_{0.1667}$	[15,24]
Mg_3Sm	*cF*16	BiF_3	*Fm-3 m*	$Mg_3(Mg,Sm)_1$	[15,22,25]
Mg_2Sm	*cF*24	Cu_2Mg	*Fd-3 m*	Laves_C15: $(Mg,Sm)_2(Mg,Sm)_1$	[15,18,22]
MgSm	*cP*2	CsCl	*Pm-3 m*	BCC_B2: $(Mg,Sm)_1(Mg,Sm)_1VA_6$	[15,16]
α-Sm	*hR*9	Sm	*R-3 m*	Rhombohedral_C19: (Mg,Sm)	[15]
β-Sm	*hP*2	Mg	*P6₃/mmc*	HCP_A3: $(Mg,Sm)_1VA_{0.5}$	[15,26,27]
γ-Sm	*cI*2	W	*Im-3 m*	BCC_A2: $(Mg,Sm)_1VA_3$	[15]

grams. The diameter of as-cast samples is about 10 mm. Subsequently, the synthesized samples were transferred to vacuum quartz tubes and subjected to prolonged annealing placed at different temperatures in a muffle furnace, followed by rapid quenching in ice water. Slight micro-segregation was present in the as-cast samples, as small pieces of pure Sm could be observed in the scanning electrom microscope (SEM) photographs of these as-cast samples. However, the micro-segregation could be eliminated by subsequent annealing treatment.

The microstructure and composition of the equilibrated and as-cast alloys were examined and analyzed using a field-emission scanning electron microscope (Hitachi SU-8000, Japan) equipped with an EDS (energy-dispersive spectroscopy) analyzer (Oxford X-MAX80, Britain). Notably, metallographic samples were initially ground with high-purity alcohol (purity: 99.7%) and then polished using an anhydrous polishing agent to prevent any reaction of Mg and Sm with water.

Phase identification of powder samples was conducted using XRD (x-ray diffractometer, Bruker D8 Discover X, Germany) with Cu-Kα radiation under low vacuum conditions. Step scanning was performed within a range of 10° ~ 90°, employing a voltage of 40 kV and a current of 40 mA.

The heating and cooling curves of selected equilibrated and as-cast samples were tested using a DSC (differential scanning colorimeter, SETARAM Labsys evo, France) at 5 °C/min. The temperature test ranges are from 300 to ~ 100 °C above the liquidus[[31]] of alloys. To avoid Mg volatilization damaging the DSC instrument or sample oxidation during DSC testing, all DSC test samples were sealed in special tantalum crucibles, which were first evacuated and then filled with high-purity argon gas.

3.2 Experimental Results and Discussion

All equilibrium and as-cast samples underwent investigation using the SEM/EDS method, with only partial samples being examined by XRD and DSC methods. Table 2 summarizes the experimental results of EDS, XRD, and DSC for all equilibrium alloys and selected as-cast samples. Annealed samples are denoted by E*i*, and as-cast samples are represented by C_i. The composition of each sample corresponds to the EDS surface scan data of the alloy. The composition of a phase within an alloy represents the average of multiple EDS data points for that particular phase.

The thermal effect temperatures of alloys, excluding the liquidus, are mainly derived from their heating curves. Table 2 presents the experimental results of liquidus for both the heating and cooling curves. DSC data without any label originates from the heating curve, while DSC data with the label "c" corresponds to the cooling curve. It is noteworthy that an anomaly was found in the present study, i.e., the liquidus temperature of Mg-Sm alloys during cooling is always slightly higher than the liquidus temperature measured on heating.

The experimental phase diagram, constructed from the experimental data of this work, is presented in Fig. 2. Micrographs of selected equilibrium or as-cast samples are shown in Fig. 3, while Fig. 4 displays the XRD patterns of alloys E3b, E7, E13a and E13b. Additionally, Fig. 5 illustrates the DSC curves for six selected alloys. A detailed discussion of these figures follows below.

3.2.1 Liquidus

Given that the liquidus temperatures derived from the heating curves are lower than those from the cooling curves in this study, the liquidus temperatures for the experimental phase diagrams presented in Fig. 2 is taken as the average of these two values.

3.2.2 $Mg_{41}Sm_5$

The existence of the phase with $Mg_{41}Ce_5$ (*I4/m*, *tI*92) type structure is confirmed with a composition close to $Mg_{89}Sm_{11}$. The micrograph of as-cast alloy C1 (Fig. 3a) provides evidence that $Mg_{41}Sm_5$ forms through peritectic reaction L + Mg_5Sm→ $Mg_{41}Sm_5$. The DSC heating curve of alloy E4 ($Mg_{83.3}Sm_{16.7}$, Fig. 5a) indicates that the peritectic reaction occurs at 568 °C. This value is between the experimental data of 570 °C reported by Zheng et al.[20] and 560 °C based on heating effects reported by Saccone et al.[15] A eutectic reaction L→Mg + $Mg_{41}Sm_5$ occurs between Mg and $Mg_{41}Sm_5$ with a eutectic point of approximately $Mg_{92}Sm_8$. The temperature of the eutectic reaction is 547 °C according to the DSC heating effects of alloys E1 and E2, which is close to the 550 °C reported by Zheng et al.[20] and slightly higher than the value of 540 °C from the heating effect of Saccone et al.[15]

3.2.3 Mg_5Sm

The crystal structure of Mg_5Sm has been verified to be isostructural to Mg_5Gd (*F-4₃ m*, *cF*440). Mg_5Sm exhibits no detectable solubility range, with its measured composition falling within the error range of its stoichiometric ratio. Formed by the peritectic reaction, L + Mg_3Sm→Mg_5Sm, at 583 °C, as evidenced by the heating curve of alloy E4 (Fig. 5a), this closely aligns with the reported data of 580 °C from the heating effect of Saccone et al.[15] However, the present experimental results suggest that Mg_5Sm is only stable in a high-temperature range and undergoes decomposition above 450 °C. Indicators of Mg_5Sm

Table 2 Summary of the experimental results obtained by EDS, XRD and DSC for the equilibrium and as-cast alloys of Mg-Sm system.

No.	Alloy Composition (at.% Sm)	Annealing		Phase	Phase com -position (at.% Sm)	Space group	Lattice parameter(Å)			DSC (°C)	Reaction	Figure
		T (°C)	Time (days)				a	b	c			
E1	2.9	400	30	Mg	0.5	*P6₃/mmc*	3.208(0)		5.198(0)	612, 617[c]	Liquidus	
				$Mg_{41}Sm_5$	11.0	*I4/m*	14.703(5)		10.358(3)	547	L → $Mg_{41}Sm_5$ + Mg	
				Eutectic	8.0							
E2	12.0	400	40	Mg_3Sm	22.3	*Fm-3m*	7.329(7)			582, 571[c]	Liquidus	
				$Mg_{41}Sm_5$	10.9	*I4/m*	14.666(8)		10.346(2)	547	L → $Mg_{41}Sm_5$ + Mg	
E3a	13.5	500	15	Mg_3Sm	21.8							Fig. 3(c)
				Mg_5Sm	17.3							
				$Mg_{41}Sm_5$	11.0							
E3b	13.5	450	15	Mg_3Sm	21.9	*Fm-3m*	7.336(7)					Fig. 3(d)
				$Mg_{41}Sm_5$	11.1	*I4/m*	14.749(1)		10.350(4)			Fig. 4(a)
E4	16.7	400	30	Mg_3Sm	22.2	*Fm-3m*	7.341(6)			654, 672[c]	Liquidus	Fig. 3(b)
				Mg_5Sm	16.7	*F-4₃m*	22.386(6)			583	L + Mg_3Sm → Mg_5Sm	Fig. 5(a)
				$Mg_{41}Sm_5$	10.9	*I4/m*	14.649(3)		10.316(2)	568	L + Mg_5Sm → $Mg_{41}Sm_5$	
				Eutectoid	15.4							
E5	23.6	400	30	Mg_3Sm	23.5					703, 719[c]	Liquidus	
										697	L → Mg_2Sm + Mg_3Sm	Fig. 5(b)
E6	27.6	550	15	Mg_2Sm	32.5							
				Mg_3Sm	24.9							
E7	30.7	400	40	Mg_3Sm	24.8	*Fm-3m*	7.353(5)			720, 729[c]	Liquidus	Fig. 3(f)
				Eutectoid	32.4					698	L → Mg_2Sm + Mg_3Sm	Fig. 4(b)
										424[c]	Mg_2Sm → Mg_3Sm + MgSm	Fig. 5(c)
E8	39.7	400	30	Mg_3Sm	25.0	*Fm-3m*	7.409(3)					
				MgSm	48.4	*Pm-3m*	3.838(8)					
				Eutectoid	32.4							
E9	43.6	720	10	MgSm	46.1	*Pm-3m*	3.826(9)					
				Mg_2Sm	33.0	*Fd-3m*	8.640(1)					
				Eutectic	37.0							
E10	45.9	550	15	MgSm	46.9					758, 779[c]	Liquidus	Fig. 5(d)
				Mg_2Sm	32.4					736	MgSm(disorder/order)	
										715	L → MgSm + Mg_2Sm	
E11	47.9	650	10	Mgsm	47.9							
E12	48.4	550	15	MgSm	48.4					766, 781[c]	Liquidus	
										738	MgSm(disorder/order)	

Table 2 continued

No.	Alloy Composition (at.% Sm)	Annealing		Phase	Phase com -position (at.% Sm)	Space group	Lattice parameter(Å)			DSC (°C)	Reaction	Figure
		T (°C)	Time (days)				a	b	c			
										715	L → MgSm + Mg_2Sm	
E13a	49.8	720	10	MgSm	47.9	*Pm-3m*	3.837(7)					Fig. 4(c)
E13b	49.8	750	10	MgSm	48.4	*Pm-3m*	3.838(1)					Fig. 4(c)
E14	51.0	760	5	MgSm	49.0							
				γ-Sm	74.7							
				Eutectic	55.0							
E15	53.1	650	10	MgSm	50.6							
				γ-Sm	72.0							
E16a	60.7	550	15	γ-Sm	74.7					789, 802[c]	Liquidus	Fig. 5(e)
				MgSm	50.2					744	L → MgSm + γ-Sm	
										588	γ-Sm → MgSm + β-Sm	
E16b	60.7	630	10	γ-Sm	71.7							
				MgSm	50.2							
E17a	64.5	670	10	γ-Sm	70.3							
				MgSm	50.2							
E17b	64.5	700	10	γ-Sm	68.0							
				MgSm	50.5	*Pm-3m*	3.843(7)					
E18	66.7	550	15	MgSm	50.3					848, 862[c]	Liquidus	
				Eutectoid	75.3					764	Solidus	
										588	γ-Sm → MgSm + β-Sm	
E19	76.4	650	10	γ-Sm	78.4					950, 951[c]	Liquidus	
				Eutectoid	74.8					851	Solidus	
										597	γ-Sm → MgSm + β-Sm	
E20	78.2	550	15	β-Sm	90.8					963, 972[c]	Liquidus	Fig. 5(f)
				MgSm	51.5					872	Solidus	
				Eutectoid	75.1					597	γ-Sm → MgSm + β-Sm	
E21	81.5	650	10	γ-Sm	81.4	…	…		…			
				β-Sm	90.7	*P6₃/mmc*	3.656(7)		5.851(2)			
E22	90.3	650	10	β-Sm	91.6					1040, 1036[c]	Liquidus	
				β-Sm	99.8					606	Solidus	
E23	91.4	550	15	β-Sm	91.9					838	Solidus	
				β-Sm	99.1					602	β-Sm β-Sm + γ-Sm	
C1	9.6	as-cast		Mg_3Sm	21.3	*Fm-3m*	7.462(3)					Fig. 3(a)
				Mg_5Sm	16.6	*F-4₃m*	22.375(7)					

Table 2 continued

No.	Alloy Composition (at.% Sm)	Annealing		Phase	Phase com -position (at.% Sm)	Space group	Lattice parameter(Å)			DSC (°C)	Reaction	Figure
		T (°C)	Time (days)				a	b	c			
				$Mg_{41}Sm5$	10.8	*I4/m*	14.657(8)		10.338(2)			
				Mg	0.9	*P6₃/mmc*	3.201(2)					
				Eutectic	7.7							
C2	26.4	as-cast		Mg_2Sm	30.9	*Fd-3m*	8.608(3)					Fig. 3(e)
				Mg_3Sm	24.6	*Fm-3m*	7.363(3)					
				Eutectic	25.6							
C3	34.7	as-cast		Mg_2Sm	32.4							
				MgSm	45.5							
				Eutectic	37.1							
C4	52.5	as-cast		MgSm	58.0					767, 792[c]	Liquidus	
				γ-Sm	65.1					744	L → MgSm + γ-Sm	
				Eutectoid	75.3							

decomposing were initially detected in alloy E4 ($Mg_{83.3}Sm_{16.7}$, annealing at 400 °C for 30 days), as shown in Fig. 3(b). The micrograph of E4 is interesting as it displays Mg_5Sm (grey phase), $Mg_{41}Sm_5$(dark phase) and the eutectoid decomposition structure of Mg_5Sm, Mg_3Sm (light grey phase) + $Mg_{41}Sm_5$ (dark phase). This indicates that at 400 °C, although Mg_5Sm undergoes eutectoid decomposition, the reaction proceeds slowly. This is also illustrated by the fact that no corresponding reaction peaks were detected in the DSC heating and cooling curves of the E4 alloy. To estimate the decomposition temperature of Mg_5Sm, another alloy was prepared in the same phase region with composition $Mg_{86.5}Sm_{13.5}$. The alloy was divided into two pieces and annealed separately at 500 and 450 °C for 15 days, referred to as alloys E3a and E3b in Table 2. The micrograph of alloy E3a (annealed at 500 °C for 15 days, Fig. 3c), reveals the presence of Mg_3Sm, Mg_5Sm, and $Mg_{41}Sm_5$. Although E3a is not completely equilibrium at 500 °C, as indicated by the persistent presence of the reactant phase Mg_3Sm, it demonstrates that Mg_5Sm is formed by the peritectic reaction L + Mg_3Sm→Mg_5Sm and remains stable at 500 °C. On the other hand, the SEM picture of alloy E3b (annealed at 450 °C for 15 days, Fig. 3d) shows that the alloy only contains Mg_3Sm and $Mg_{41}Sm_5$ phases. The XRD pattern of alloy E3b (Fig. 4a) confirms the presence of Mg_3Sm and $Mg_{41}Sm_5$, consistent with SEM/EDS results. Micrographs of E3a and E3b suggest that Mg_5Sm decomposes (Mg_5Sm→Mg_3Sm + $Mg_{41}Sm_5$) above 450 °C and below 500 °C.

3.2.4 Mg_3Sm

The XRD patterns of several alloys confirm that the crystal structure of Mg_3Sm is of BiF_3 type (*Fm-3 m*, *cF*16), which is the ordered structure of Bcc, commonly known as $D0_3$. The EDS data for alloys E3 to E7 show that Mg_3Sm exhibits some solubility, ranging from 22 to 25 at.% Sm.

The micrograph of as-cast alloy C2 (Fig. 3e) suggests that Mg_3Sm is congruently formed from the liquid phase instead by a peritectic reaction. Since a eutectic structure with an overall composition of $Mg_{74.4}Sm_{25.6}$ can be observed and the eutectic point is located between Mg_2Sm ($Mg_{69}Sm_{31}$) and Mg_3Sm ($Mg_{75.4}Sm_{24.6}$), it indicates that the eutectic point of L→Mg_2Sm + Mg_3Sm is at 25.6 at.% Sm. The DSC heating curve of alloy E7 and E5 (Fig. 5b) suggests the eutectic reaction occurs at 697 °C, which is close to the value of 700 °C reported by Saccone et al.[15] The melting point of Mg_3Sm should be very close to the liquidus of alloy E5 ($Mg_{76.4}Sm_{23.6}$), which is 711 ± 8 °C.

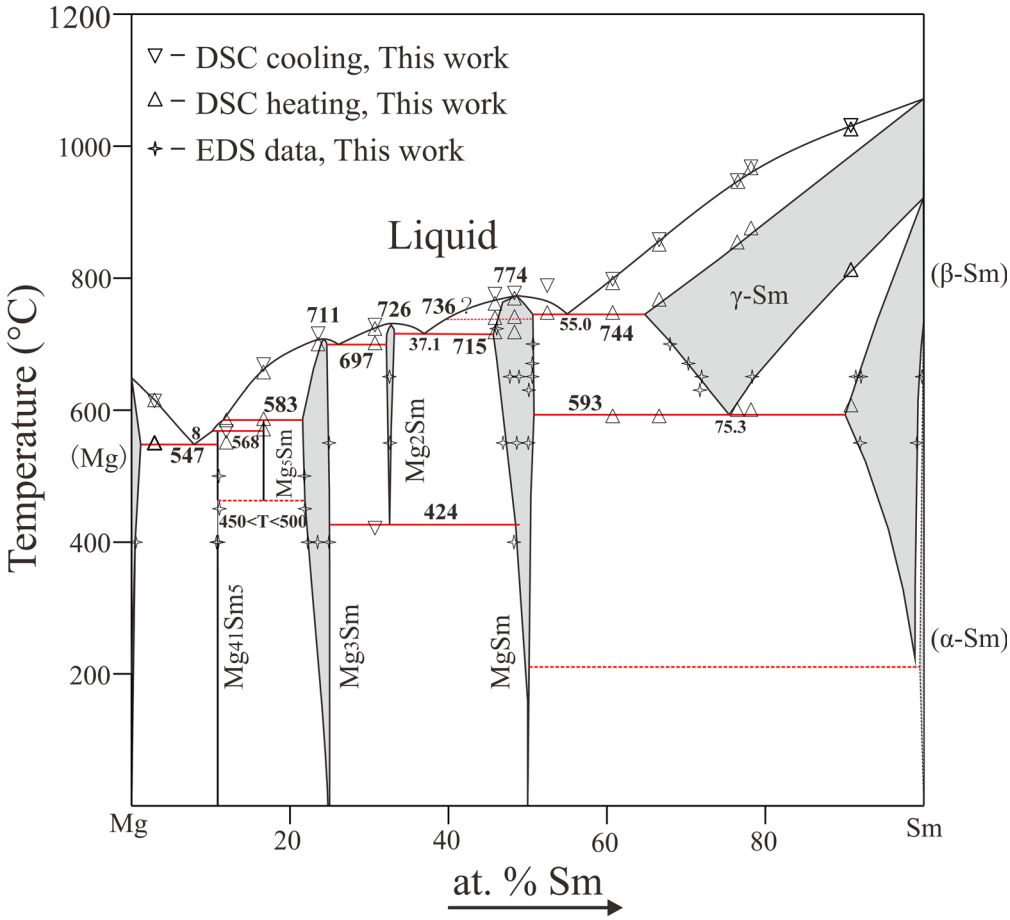


Fig. 2 The updated experimental phase diagram of Mg-Sm system from this work

3.2.5 Mg_2Sm

Mg_2Sm is confirmed to be a congruently melting phase, and its crystal structure is confirmed to be Cu_2Mg type (*Fd-3 m*, *cF*24). Judging from the measured composition of Mg_2Sm in the as-cast samples, C2 and C3, Mg_2Sm exhibits a small solid solution range of about 1.5 at.% Sm at the high temperatures. However, below 550 °C, the measured composition of Mg_2Sm is 32.4 at.% Sm, without a detectable solution range. Between Mg_2Sm and MgSm, a eutectic reaction (Liquid→MgSm + Mg_2Sm) occurs at 37.1 at.% Sm and 715 °C. The eutectic point composition was determined in the SEM/EDS results of the C3 sample, and the eutectic reaction temperatures were determined in the heating curves of alloys E10 and E12. The reaction temperature is 20 °C lower than the value of 735 °C reported by Saccone et al.[15] In this study, Mg_2Sm was found to exist stably only at high-temperatures, and it decomposes at 424 °C through the eutectoid reaction Mg_2Sm→ MgSm + Mg_3Sm. Relevant experimental evidence is presented in Figs. 3(f), 4(b) and 5(c). Figure 3(f) displays the micrograph of the equilibrium alloy E7 ($Mg_{69.3}Sm_{30.7}$) annealed at 400 °C for 40 days. The image reveals that the alloy consists of a Mg_3Sm phase and a eutectoid decomposition structure with an overall composition of $Mg_{67.6}Sm_{32.4}$. This indicates that the Mg_2Sm has undergone eutectoid decomposition above 400 °C. The XRD pattern of alloy E7 (Fig. 4b) shows that the alloy consists of Mg_3Sm and MgSm two phases, further indicating that Mg_2Sm decomposes into Mg_3Sm and MgSm phases. The DSC cooling curve of alloy E7 (Fig. 5c) exhibited a weak peak at 424 °C, but no corresponding peak was detected in the heating curve of E7. The 424 °C is inferred as the reaction temperature for this eutectoid reaction (Mg_2Sm→Mg_3Sm + MgSm). We did not directly determine the melting point of Mg_2Sm, but inferred from the liquidus temperature of 726 ± 6 °C for alloy E7($Mg_{69.3}Sm_{30.7}$), suggesting that the melting point of Mg_2Sm should be slightly higher than 726 °C.

3.2.6 MgSm

MgSm forms congruently from liquid at about 774 ± 8 °C a temperature about 26 °C lower than 800 ± 10 °C

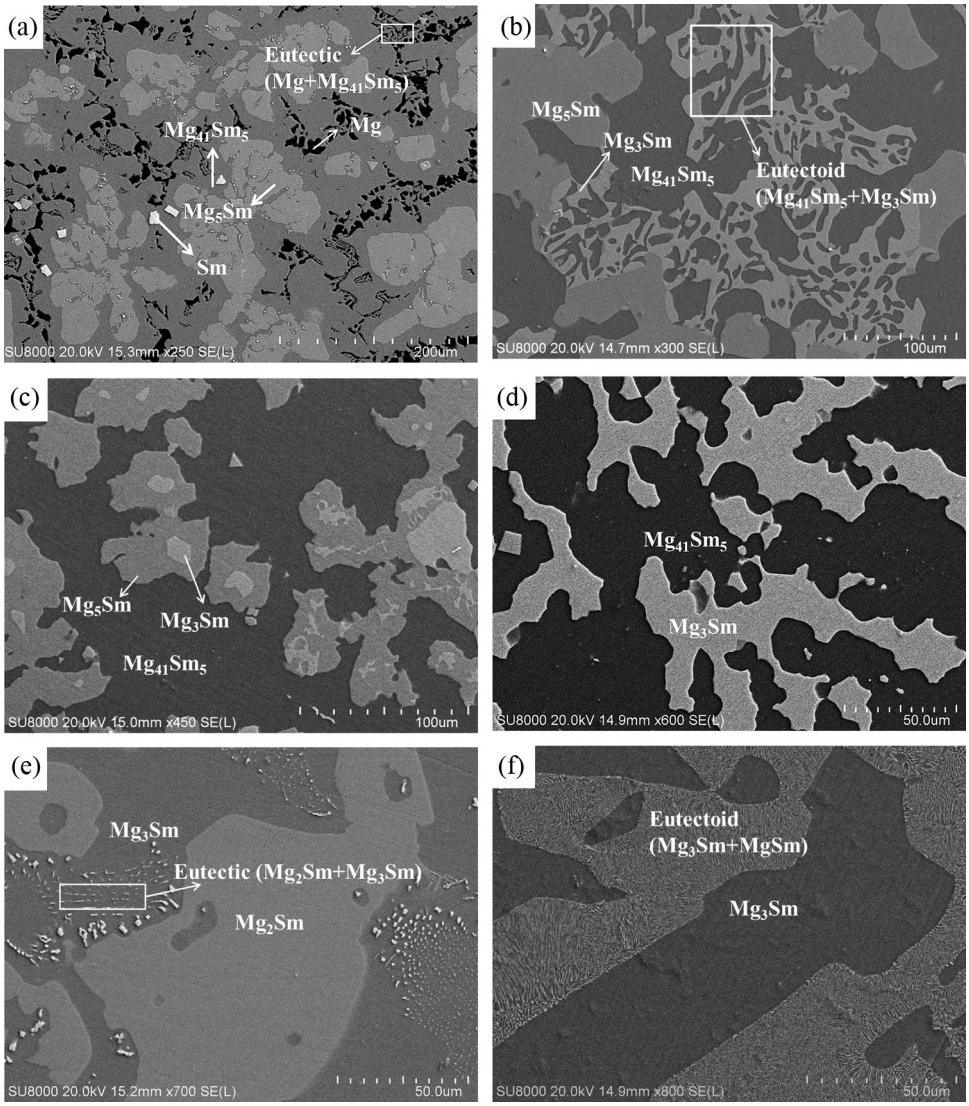
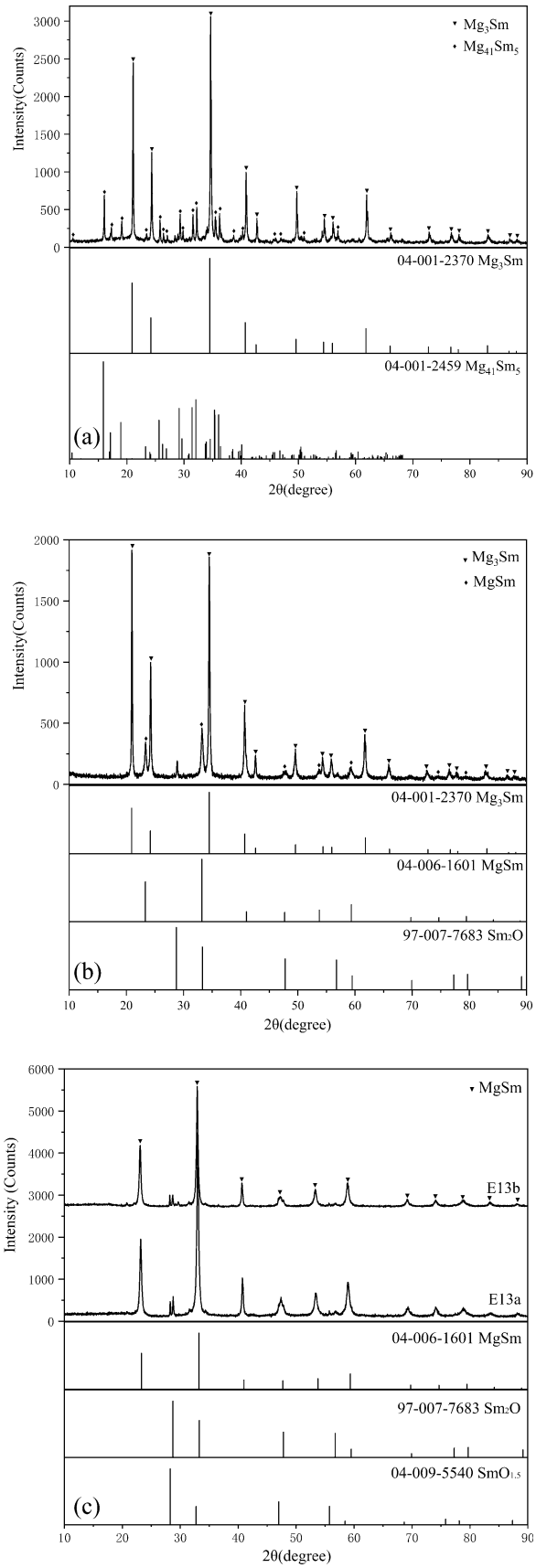


Fig. 3 SEM pictures of equilibrium or as-cast alloys: (a) as-cast alloy C1 ($Mg_{90.4}Sm_{9.6}$) showing the primary phase Mg_5Sm (light gray) surrounded by $Mg_{41}Sm_5$ (gray phase) and eutectic structure ($Mg + Mg_{41}Sm_5$); (b) equilibrium alloy E4 ($Mg_{83.3}Sm_{16.7}$, annealed at 400 °C for 30 days) showing coexistence of Mg_5Sm (gray phase), $Mg_{41}Sm_5$ (dark) and eutectoidally decomposed structure Mg_3Sm (light gray) + $Mg_{41}Sm_5$; (c) annealed alloy E3a ($Mg_{86.5}Sm_{13.5}$, annealed at 500 °C for 15 days) shows that $Mg_{41}Sm_5$ (black) is in equilibrium with Mg_5Sm (gray), and that the upper phase Mg_3Sm (light gray)surrounded by Mg_5Sm; (d) equilibrium alloy E3b ($Mg_{86.5}Sm_{13.5}$, annealed at 450 °C for 15 days) showing coexistence of Mg_3Sm (gray) + $Mg_{41}Sm_5$ (black); (e) as-cast alloy C2 ($Mg_{73.6}$-$Sn_{26.4}$) showing the primary phase Mg_2Sm ($Mg_{69}Sm_{31}$, gray), surrounded by secondary phase Mg_3Sm ($Mg_{74.4}Sm_{24.6}$, dark) and eutectic structure, $Mg_3Sm + Mg_2Sm$ ($Mg_{74.4}Sm_{25.6}$); (f) equilibrium alloy E7 ($Mg_{69.3}Sm_{30.7}$, annealed at 400 °C for 40 days) showing the presence of Mg_3Sm (dark) and a eutectoid decomposition structure ($Mg_3Sm + MgSm$, $Mg_{77.6}Sm_{32.4}$)

reported by Saccone et al.[15] The maximum solid solubility range for MgSm is about 5 at.% Sm at 715 °C, decreasing to 2 at.% Sm at 400 °C. The crystal structure of MgSm at 400 °C is confirmed to be of the CsCl type (*Pm-3 m*, *cP*2), representing a Bcc (Body-Centered Cubic) ordered structure, specifically the Bcc_B2. In the DSC heating curves of sample E10 ($Mg_{54.1}Sm_{45.9}$, Fig. 5d), three thermal effects were detected. The onset point of the peak at 715 °C is inferred to be the eutectic reaction temperature of L→MgSm + Mg_2Sm. The peak temperature of the highest temperature peak (758 °C) is considered as the liquidus of the alloy. Between these two thermal effects, a small peak with an onset of 736 °C is also detected in alloy E12 ($Mg_{51.6}Sm_{48.4}$). We hypothesized that MgSm might undergo a disorder/order (Bcc-A2/BCC-B2) transition at 737 ± 1 °C. In order to test this hypothesis, we prepared an alloy with composition $Mg_{50.2}Sm_{49.8}$, divided it into two pieces, annealed separately at 720 °C (E13a) and 750 °C (E13b) for 10 days, and then measured their XRD patterns. Fig. 4(c) shows the XRD pattern of E13a and E13b, from



◀**Fig. 4** X-ray diffraction patterns obtained from equilibrium alloys: (a) E3b ($Mg_{86.5}Sm_{13.5}$, annealed at 450 °C for 15 days) showing coexistence of Mg_3Sm and $Mg_{41}Sm_5$; (b) E7 ($Mg_{69.3}Sm_{30.7}$, annealed at 400 °C for 40 days) shows that the alloy contains Mg_3Sm, MgSm and small amount SmO_2; (c) the XRD patterns of E13a ($Mg_{50.2}Sm_{49.8}$, annealed at 750 for 10 days) and E13b ($Mg_{50.2}Sm_{49.8}$, annealed at 720 for 10 days) shows that the phases in E13a and E13b have the same structure MgSm (*Pm-3m*, *cP*2)

which it can be seen that these two alloys have the same structure, i.e., the CsCl-type structure (BCC-B2), and this experimental result does not support our conjecture. The phase transition associated with the thermal effect at 736 °C remains to be determined experimentally.

3.2.7 Sm

At varying temperatures, the element Sm exists as three different allotropic phases, namely, α-Sm, *β*-Sm and *γ*-Sm. The crystal structure of *β*-Sm is confirmed as Mg type (*P6₃/mmc*, *hP*2). However, the X-ray diffraction (XRD) patterns obtained in this study did not confirm the crystal structure of *γ*-Sm. The reason for this is the apparent preferential orientation observed in the XRD patterns of the powder samples containing the *γ*-Sm phase, which is attributed to the inherent ductility of *γ*-Sm. It is estimated that there are three invariant reactions between the Sm allotropes and its neighboring binary compound MgSm. The highest temperature invariant reaction between MgSm and Sm should be a eutectic reaction L→MgSm + (*γ*-Sm) since MgSm forms congruently from liquid, and the eutectic point was detected as $Mg_{45}Sm_{55}$ in alloy E14. Based on the DSC heating curve for samples E16a (Fig. 5e) and C4, the temperature for the invariant reaction is 744 °C. This value is 41 °C lower than the 785 °C reported by Saccone et al.[15] The intermediate temperature invariant reaction between Sm and MgSm is the eutectoid reaction (*γ*-Sm) → MgSm + (*β*-Sm). The eutectoid decomposition structure of *γ*-Sm can be observed in several alloys, and the eutectoid point was detected at about 75.3 at.% Sm. This reaction temperature was detected in the DSC heating curves of 4 samples (E16a, E18, E19 and E20). Samples E16a (Fig. 5e) and E18 have measurements of 588 °C, and samples E19 and E20 (Fig. 5f) have measurements of 597 °C. The average of these values was regarded as the reaction temperature for this equilibrium reaction (*γ*-Sm) →MgSm + (*β*-Sm), i.e. 593 ± 5 °C. The value is 13 °C higher than the literature data of Saccone et al.[15] The lowest temperature invariant reaction between Sm and MgSm is suggested to be the eutectoid reaction: (*β*-Sm) →MgSm + (α-Sm). However, this equilibrium reaction temperature was not detected in our present DSC experiments.

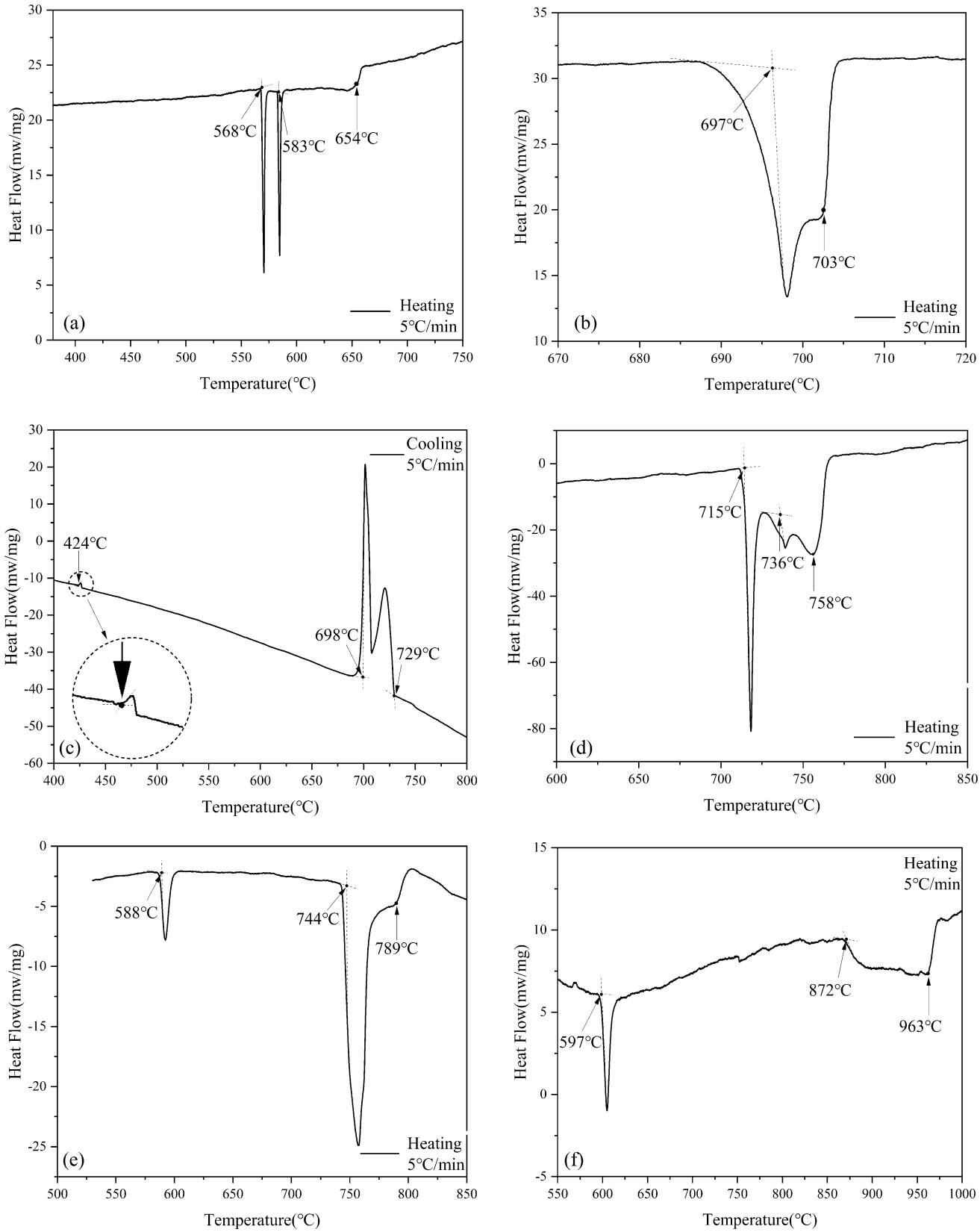


Fig. 5 DSC heating or cooling curves for selected Mg-Sm alloys: (a) E4 ($Mg_{83.3}Sm_{16.7}$); (b) E5 ($Mg_{76.4}Sm_{23.6}$); (c) E7 ($Mg_{69.3}Sm_{30.7}$); (d) E10($Mg_{54.1}Sm_{45.9}$). (e) E16a ($Mg_{29.3}Sm_{60.7}$), (f) E20 ($Mg_{21.8}Sm_{78.2}$)

The solution ranges of the phases in the Mg-Sm phase diagram reported by Saccone et al.[15] were estimated from earlier experimental reports.[18,21] In order to determine the boundary between the (γ-Sm) and MgSm phases, several samples located in the (γ-Sm) + MgSm two-phase region were prepared and divided into two or three pieces, which were annealed at different temperatures, and then the compositions of both phases in these alloys were examined by SEM/EDS.

Comparison of Figs. 1 and 2 shows that there are four main differences between our experimental phase diagrams and those of Saccone et al.[15]: (i) the Mg_5Sm and Mg_2Sm phases exist only at high temperatures, and they decompose above 400 °C; (ii) Mg_3Sm is congruently formed from liquid ra-ther than from a peritectic reaction; (iii) in the range of 33.0 ~ 70.0 at.% Sm, there is a relatively large difference between the phase transition temperatures of Saccone et al.[15] and our experimental data; and iv) the solid solution ranges of (Mg), Mg_3Sm, Mg_2Sm, MgSm, γ-Sm and β-Sm were dedicatedly determined in the present work.

4 Thermodynamic Assessment

4.1 Thermodynamic Model

The Gibbs free energy of pure elements was obtained from the SGTE database compiled by Dinsdale.[33] The thermodynamic models employed for each phase in this work are summarized in Table 1 and are discussed in detail below.

4.1.1 Solution Phase

Liquid, Hcp_A3 (Mg and β-Sm), Bcc_A2 (γ-Sm) and Rhombohedral_C19 (α-Sm), were treated as disordered substitutional solutions. The Gibbs energy for a given phase φ is expressed as:

$$G_m^{\varphi} = (x_{Mg}^0 G_{Mg}^{\varphi} + x_{Sm}^0 G_{Sm}^{\varphi}) + RT\left(x_{Mg}\ln_{Mg} + x_{Sm}\ln_{Sm}\right) + {}^{E}G_m^{\varphi} \quad \text{(Eq 1)}$$

where $\left(x_{Mg}0G_{Mg}^{\varphi} + x_{Sm}0G_{Sm}^{\varphi}\right)$ is the Gibbs energy of an unreacted mixture of Mg and Sm of φ phase; $RT\left(x_{Mg}\ln_{Mg} + x_{Sm}\ln_{Sm}\right)$ is Gibbs energy contribution from the configurational entropy of the phase; ${}^{E}G_m^{\varphi}$ is the excess Gibbs energy, and is expressed by the Redlich–Kister[34] polynomial:

$${}^{E}G_m^{\varphi} = x_{Mg}x_{Sm}\sum_{i=0}^{n} {}^{i}L^{\varphi}(x_{Mg} - x_{Sm})^i \quad \text{(Eq 2)}$$

where ${}^{i}L^{\varphi}$ are the interaction parameters between elements Mg and Sm, and their general form is expressed as :

$${}^{i}L^{\varphi} = a + bT \quad \text{(Eq 3)}$$

The "a" and "b" are the parameters to be optimized during the optimization process.

4.1.2 Stoichiometric Compounds

The $Mg_{41}Sm_5$ and Mg_5Sm with negligible solubility are described as stoichiometric compounds. The Gibbs energy per mole of the formula unit of Mg_xSm_y is expressed using the Neumann-Kopp[35] rule as:

$$G_m^{Mg_xSm_y} - xH_{Mg}^{SER} - yH_{Sm}^{SER} = a + bT + x^0G_{Mg}^{SER} + y^0G_{Sm}^{SER} \quad \text{(Eq 4)}$$

where a and b are the parameters to be optimized in the optimization process.

4.1.3 Sublattice Models for Mg_2Sm, MgSm and Mg_3Sm

Mg_2Sm was determined to have about 1.5 at.% Sm solubility range, we adopted the convenient two-sublattice model of Laves_C15 to describe Mg_2Sm ($cF24$, Cu_2Mg) as (**Mg**, Sm)$_2$(Mg, **Sm**). The Gibbs energy of Mg_2Sm can be expressed as the follows[36]:

$$\begin{aligned} G_m^{Mg_2Sm} =& y'_{Mg}y''_{Mg}G_{Mg:Mg}^{Mg_2Sm} + y'_{Mg}y''_{Sm}G_{Mg:Sm}^{Mg_2Sm} \\ &+ y'_{Sm}y''_{Mg}G_{Sm:Mg}^{Mg_2Sm} + y'_{Sm}y''_{Sm}G_{Sm:Sm}^{Mg_2Sm} \\ &+ 2RT\left(y'_{Mg}\ln y'_{Mg} + y'_{Sm}\ln y'_{Sm}\right) \\ &+ RT\left(y''_{Mg}\ln y''_{Mg} + y''_{Sm}\ln y''_{Sm}\right) \\ &+ \sum_{i1}\sum_{i2}\sum_{k} y'_{i1}y'_{i2}y''_{k}L_{i1,i2:k}^{Mg_2Sm} \\ &+ \sum_{k}\sum_{i1}\sum_{i2} y'_{k}y''_{i1}y''_{i2}L_{k:i1,i2}^{Mg_2Sm} \end{aligned} \quad \text{(Eq 5)}$$

Where y'_i and y''_j are the site fractions of the constituents Mg or Sm on the first and second sublattice.$G_{i:j}^{Mg_2Sm}$ represents the Gibbs energy of the end-member with i_2j. $L_{i1,i2:k}^{Mg_2Sm}$, and $L_{k:i1,i2}^{Mg_2Sm}$ represent the binary interaction parameters.

Ithis optimization, the γ-Sm ($cI2$, W, Bcc_A2), the MgSm ($cP2$, CsCl, Bcc_B2) and the Mg_3Sm ($cF16$, BiF_3, $D0_3$) are described by three separate models. MgSm (Bcc_B2), with maximum solid solubility of 5 at.% Sm was described by a sublattice model as $(Mg,Sm)_1$(-$Mg,Sm)_1Va_6$. Mg_3Sm was determined to have about 3 at.%

Table 3 Thermodynamic parameters in the Mg-Sm binary system optimized in the present work.

Phase	Thermodynamic model	Parameters(J/mol)
Liquid	(Mg, Sm)	$G^{Liq}_{Mg} = GLIQMG$
		$G^{Liq}_{Sm} = GLIQSM$
		${}^{0}L^{Liq}_{Mg,Sm} = -50894 + 20.34*T$
		${}^{1}L^{Liq}_{Mg,Sm} = -29763 + 12.5*T$
BCC_A2	$(Mg, Sm)_1VA_3$	$G^{BCC_A2}_{Mg:VA} = GBCCMG$
		$G^{BCC_A2}_{Sm:VA} = GBCCSM$
		${}^{0}L^{BCC_A2}_{Mg,Sm:VA} = -47266 + 19.2*T$
		${}^{1}L^{BCC_A2}_{Mg,Sm:VA} = -34770 + 19.4*T$
BCC_B2	$(Mg, Sm)_1 (Mg, Sm)_1 VA_6$	$G^{BCC_B2}_{Mg:Mg:VA}$=2GBCCMG
		$G^{BCC_B2}_{Sm:Sm:VA}$=2GBCCSM
		$G^{BCC_B2}_{Sm:Mg:VA}=G^{BCC_B2}_{Mg:Sm:VA}$=GHSERMG+GHSERSM-35017+9.24*T
		${}^{0}L^{BCC_B2}_{Mg:Mg,Sm:VA} = {}^{0}L^{BCC_B2}_{Mg,Sm:Mg:VA}$=15600-24*$T$
		${}^{0}L^{BCC_B2}_{Mg,Sm:Sm:VA} = {}^{0}L^{BCC_B2}_{Sm:Mg,Sm:VA}$=27000
HCP_A3	$(Mg, Sm)_1VA_3$	$G^{HCP_A3}_{Mg:VA} = GHSERMG$
		$G^{HCP_A3}_{Sm:VA} = GHCPSM$
		${}^{0}L^{HCP_A3}_{Mg,Sm:VA} = -22000 + 6.7*T$
		${}^{1}L^{HCP_A3}_{Mg,Sm:VA} = -6800$
Rhombohedral_C19	(Mg, Sm)	$G^{Rho_C19}_{Sm} = GHSERSM$
		${}^{0}L^{Rho_C19}_{Mg,Sm} = -20000$
Laves_C15	$(Mg, Sm)_2(Mg, Sm)_1$	$G^{Laves_C15}_{Mg:Mg}$=GL-C15MG
		$G^{Laves_C15}_{Sm:Sm}$=3GHSERSM + 24000
		$G^{Laves_C15}_{Sm:Mg}$=2GHSERSM+GHSERMG+15000
		$G^{Laves_{C15}}_{Mg:Sm}$=2GHSERMG+GHSERSM-40184+1.57*T
Mg_3Sm	$Mg_3(Mg, Sm)_1$	$G^{Mg_3Sm}_{Mg:Mg}$=4GHSERMG + 24000
		$G^{Mg_3Sm}_{Mg:Sm}$=3GHSERMG+GHSERSM-55903+9.162*T
		${}^{0}L^{Mg_3Sm}_{Mg:Mg,Sm} = -17880-4.0*T$
Mg_5Sm	$Mg_{0.8333}Sm_{0.1667}$	$G^{Mg_5Sm}_{Mg:Sm}$=0.8333GHSERMG + 0.16667GHSERSM − 8654.5-0.12*T
$Mg_{41}Sm_5$	$Mg_{0.8913}Sm_{0.1087}$	$G^{Mg_{41}Sm_5}_{Mg:Sm}$=0.8913GHSERMG + 0.1087GHSERSM-6068+0.02*T

GL-C15MG is taken from the Pure 5 database of Thermo-Calc software.

Sm solubility range and was described by a sublattice model of Mg_3(Sm, Mg) by Kang et al..[31] The model was accepted in the present work. The Gibbs energy of Mg_3Sm and MgSm are defined analogous to Eq. 5.

4.2 Calculated Results and Comparison with Experiments

Based on the present experimental data and reliable literature data, the Mg-Sm binary phase diagram was re-optimized thermodynamically using Thermo-Calc software. The thermodynamic parameters of the phases obtained by the optimization are shown in Table 3. The invariant equilibrium data obtained from the calculations are summarized in Table 4, where the experimental data are also listed for comparison. Comparison of the calculated results with the experimental data will be discussed subsequently.

Figure 6(a) provides the calculated phase diagram with experimental data from this work and literature.[15] The difference between our experimental data and the literature data[15] can be seen in Fig. 6(a). In the range of 0 ~ 33.3 at.% Sm, the DTA heating data of Saccone et al.[15] are not

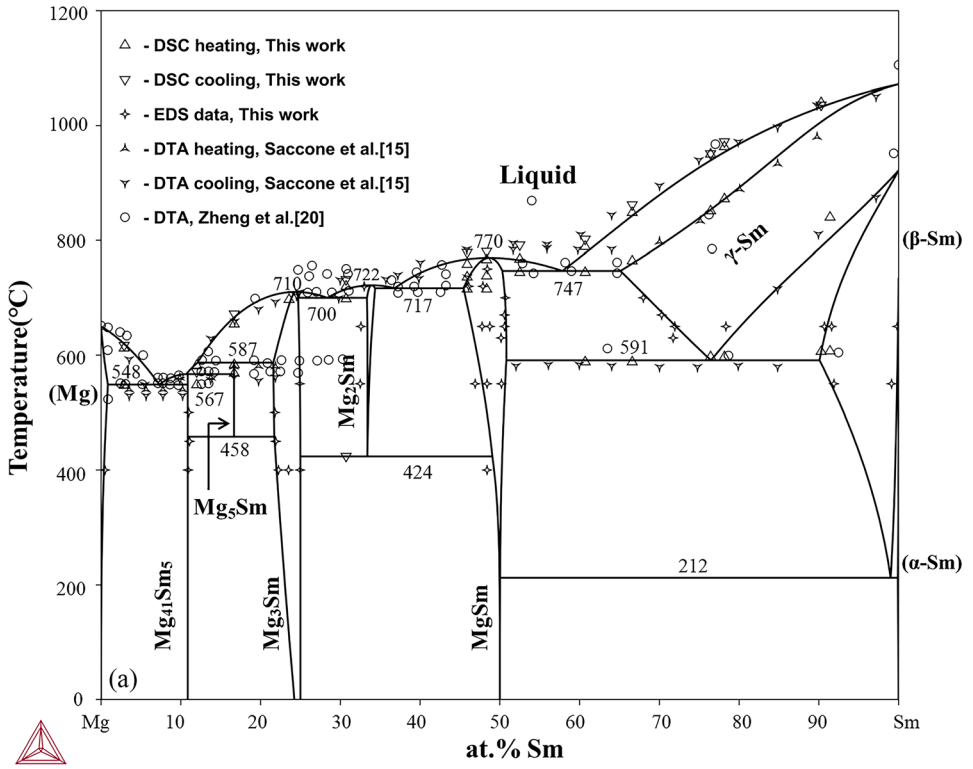
Table 4 Comparison of calculated and experimental data for invariant reactions in Mg-Sm binary system.

Invariant reaction	Reaction type	Composition (at.%Sm)			T (°C)	References
L⇆ Mg + $Mg_{41}Sm_5$	Eutectic	7.3	0.9	10.9	548	Cal. TW
		8.0	0.9	10.9	547	TW
		8.0	…	10.9	530	Cooling-[[15]]
		8.0	…	10.9	540	Heating-[[15]]
		…	1.0	10.9	542	Ref 18
L + Mg_5Sm⇆$Mg_{41}Sm_5$	Peritectic	10.7	16.7	10.9	567	Cal. TW
		…	16.7	10.9	568	TW
		…	16.7	10.9	560	Heating-[[15]]
L + Mg_3Sm⇆Mg_5Sm	Peritectic	12.1	21.6	16.7	587	Cal. TW
		…	21.2	16.7	583	TW
		10.0	25.0	16.7	565	Cooling-[[15]]
		10.0	25.0	16.7	580	Heating-[[15]]
Mg_5Sm⇆$Mg_{41}Sm_5$ + Mg_3Sm	Eutectoid	16.7	10.9	21.7	458	Cal. TW
		16.7	10.9	21.9	450-500	TW
L⇆Mg_3Sm	Congruent	24.4			710	Cal. TW
		25.0				TW
L⇆Mg_2Sm + Mg_3Sm	Eutectic	28.4	33.3	24.7	700	Cal. TW
		25.6	30.9	25.0	697	TW
L⇆Mg_2Sm	Congruent	33.9			722	Cal. TW
		32.4			730	TW
		33.3			750	Cooling-[[15]]
L⇆Mg_2Sm + MgSm	Eutectic	36.8	34.4	45.5	717	Cal. TW
		37.1	33.3	45.5	715	TW
		35.5	33.3	50.0	735	Cooling-[[15]]
		37.9	33.3	50.0	720	Ref 20
Mg_2Sm⇆Mg_3Sm + MgSm	Eutectoid	33.4	25.0	49.1	424	Cal. TW
		32.4	25.0	48.0	424	TW
L⇆MgSm	Congruent	48.6			770	Cal. TW
		48.0			774	TW
					800	Cooling-[[15]]]
L⇆MgSm + γ-Sm	Eutectic	57.9	50.3	65.1	747	Cal. TW
		55.0	50.8	64.8	744	TW
		55.0	50.0	66.0	785	Cooling-[[15]]
γ-Sm⇆MgSm + β-Sm	Eutectoid	76.5	50.8	90.1	591	Cal. TW
		75.3	50.2	…	593	TW
		80.0	50.0	…	580	Cooling-[[15]]
β-Sm⇆MgSm + α-Sm	Eutectoid	99.0	50.0	99.9	212	Cal. TW

TW: This work.

significantly different from our present experimental results. In the range of 33.3 ~ 65.0 at.% Sm, the experimental data of the present work differs significantly from that of Saccone et al.[15] and Zheng et al.[20] The present measured reaction temperatures of invariant reactions L→MgSm + Mg_2Sm and L→MgSn + (γ-Sm) are 715 and 744 °C, which are 20 and 41 °C lower than the experimental values of 735 and 785 °C reported by Saccone et al.,[[15]] respectively. These two equilibrium reaction temperatures are computed as 717 °C and 747 °C in present assessment, and the difference from the current experimental values is less than 3 °C. In the range of 70.0 ~ 100 at.% Sm, the liquidus and solidus of Saccone et al.[15] are in good agreement with our experimental values, and well fitted by the present description.

The present optimization has put a high weight on the experimental results of this work, so the calculated phase diagram fits well with our experimental results (as shown in Fig. 6b). The decomposition temperatures of Mg_2Sm and Mg_5Sm were calculated to be 425 and 458 °C, respectively, in good agreement with present experimental data. The calculated values for the solid solution ranges of



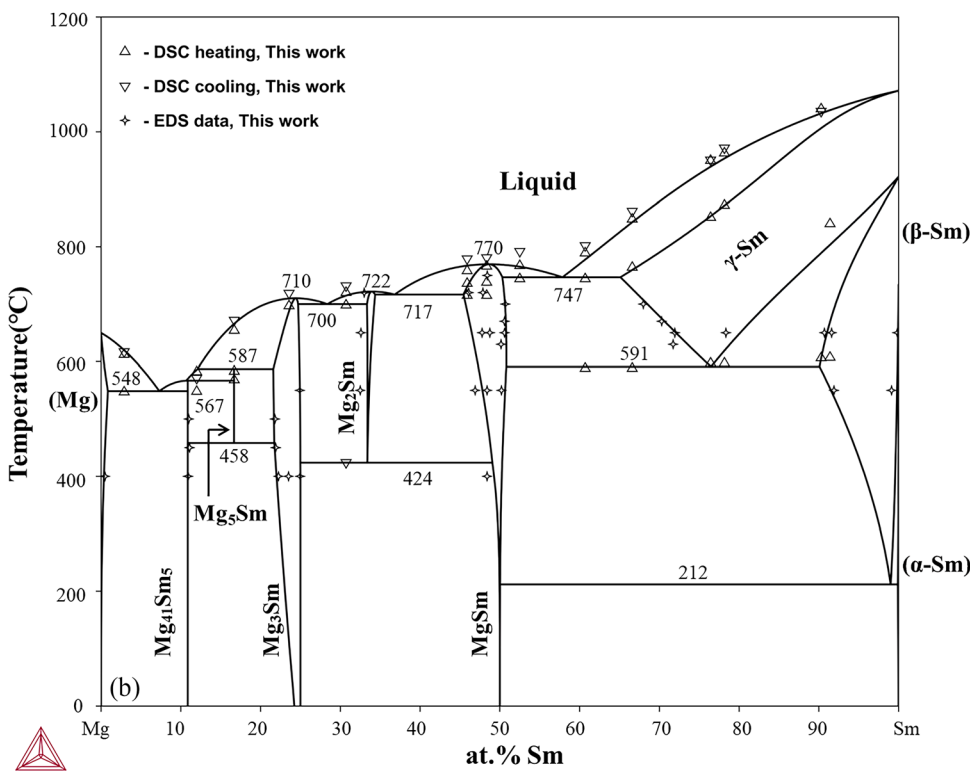


Fig. 6 The calculated phase diagram of Mg-Sm binary system based on current thermodynamic description, (a) comparison with experimental data of this work and literature data [[15,20]]; (b) comparison with the experimental data of this work.

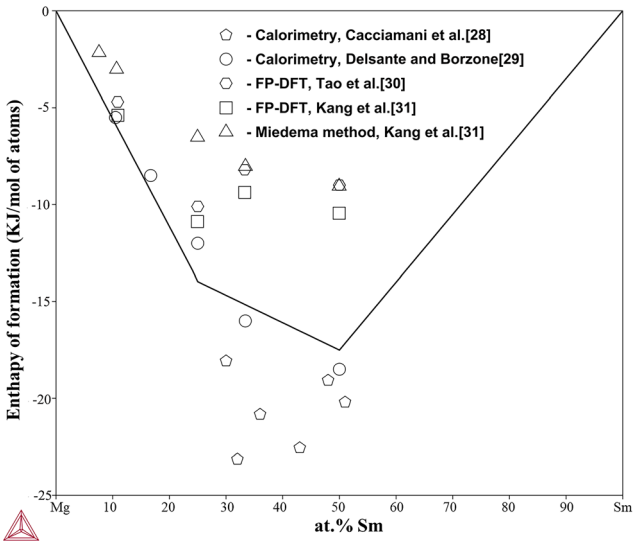


Fig. 7 Calculated standard enthalpies of formation at 300 K compared with the literature data[[28–31]]

Mg_3Sm, MgSm, and γ-Sm are in good agreement with the experimental values. In addition, the thermodynamic parameters of Hcp_A3 characterize the solid solution ranges of (Mg) and β-Sm well. The current calculated eutectoid position of Mg_2Sm is located near 33.3 at.% Sm, dictated by its thermodynamic model, which deviates from the experimental value of 32.4 at.% Sm. All calculated invariant equilibrium temperatures were within 5 °C with the experimental values of this work. The calculated melting points of Mg_3Sm, Mg_2Sm and MgSm are 710, 722 and 770 °C, respectively, which match the experimental data of this work well. The agreement between the calculated liquidus and experimental values is also within the error of the experiments.

The calculated enthalpies of formation of the compounds at 300 K are shown in Fig. 7, for comparison with the calorimetric data of Cacciamani et al.[28] and Delsante and Borzone[29] as well as calculated data of Kang et al.[31] and Tao et al.[30] The experimental data[28,29] are more negative than the calculated data.[30,31] In addition, there is a large experimental error in the early calorimetric data of Cacciamani et al..[28] The present optimization has put a high weight to the experimental data of Delsante and Borzone.[29] The calculated results are in good agreement with the experimental data.[29]

5 Summary and Conclusions

An updated experimental phase diagram of the Mg-Sm system is presented based on the experimental data of this work. A new comprehensive thermodynamic reassessment of the Mg-Sm binary system has been carried out based on experimental data of this work and reliable literature data. The following conclusions can be drawn from this study:

1. The existence of five binary compounds, $Mg_{41}Sm_5$, Mg_5Sm, Mg_3Sm, Mg_2Sm and MgSm, in the Mg-Sm system was confirmed. However, Mg_2Sm and Mg_5Sm are only stable in the high-temperature region, Mg_2Sm decomposes at 424 °C and Mg_5Sm decomposes at a temperature in the range of 450 ~ 500 °C.
2. Mg_3Sm is formed congruently from the liquid rather than through the peritectic reaction $L + Mg_2Sm \rightarrow Mg_3Sm$.
3. Solution regions for Mg_3Sm, Mg_2Sm, MgSm and (γ-Sm) were identified.
4. All invariant reaction temperatures except the one of (β-Sm) → (α-Sm) + MgSm were updated.
5. A comprehensive reevaluation of the Mg-Sm binary system was conducted using the Calphad method, taking into account the new experimental data and reliable thermodynamic data. Solution phases MgSm, Mg_2Sm and Mg_3Sm were described with the sublattice model. The optimized thermodynamic parameters successfully describe both literature data and the newly acquired experimental results. All calculated invariant reaction temperatures, liquidus and solidus, agree well with the experimental data. This work contributes to a more accurate understanding of the Mg-Sm system's phase behavior and thermodynamic properties.

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